Students in need of free code to access the ebook should email me by next Monday (03/03/25).

Today

AGENDA

Short review of propagation error

Linear Combinations of Measurements

Point Estimation

SCHEDULE

LearnHub Indr252:

- Reading assignment 1 due next Friday (07/03/25)
- Class exercises and their solutions

LearnHub Indr252 LabA

- Lab 2 today at 17:30 in SNAB149: assignment due at 19:00 today (no attendance only submissions)
- Lab videos will be placed in LearnHub: INDR252-Spring25-LABA-CE → General → Lab Videos

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Parts of Error

We can think of the error of the measurement as being composed of two parts:

- **Systematic error (bias) - accuracy**
  Bias is the part of the error that is the same for every measurement. For example, a scale that always gives you a reading that is too low.

- **Random error - precision**
  Random error is error that varies from measurement to measurement and averages out to zero in the long run.

Measured value = true value + bias + random error

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Bias

sistematik hata

yanlılık

eğilim

önyargı

taraf tutma

eğimli

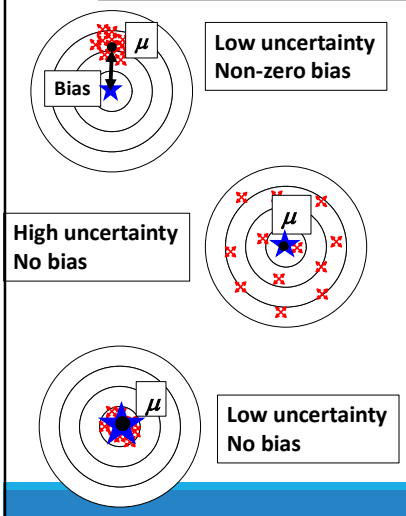
çapraz

meyil

- **unbiased estimator:** yansız kestirici
- **biased estimator:** yanlı kestirici

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Propagation of error



Target → True value ★

Shots → Measurements ✕

Average of shots → Mean measurement (μ) •

Spread of shots → Uncertainty (σ)

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Concepts to be repeated in different contexts

We will see the same concepts in point estimation as well.

- **Systematic error (bias) - accuracy**
- **Random error - precision**

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Linear Functions of Measurements

X: a random variable, $\mu = E(X)$ and $\sigma^2 = \mathrm{Var}(X)$

$Y = a + bX$: function of X

$E(Y)$ and $\mathrm{Var}(Y)$: ?

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a random sample with common mean μ and variance σ^2

$Y = \sum_{i=1}^{n} c_i X_i$: function of $(X_1, X_2, \cdots, X_n)$, where c_i's are constants

$E(Y)$ and $\mathrm{Var}(Y)$: ?

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Example

A surveyor is measuring the perimeter of a rectangular lot. He measures two adjacent sides to be 50.11 ± 0.05m and 75.12 ± 0.08m. These measurements are independent. Estimate the perimeter of the lot and find the uncertainty in the estimate.

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a random sample with common mean μ and variance σ^2

Sample mean:

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n}$$

$E(\bar{X})$ and $\mathrm{Var}(\bar{X})$: ?

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Example

The length of a component is to be measured by a process whose uncertainty is 0.05 cm. If 25 independent measurements are made and the average of these is used to estimate the length, what will the uncertainty be? How much more precise is the average of 25 measurements than a single measurement?

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Linear Combinations of Measurements

X: a measurement with a certain device with mean μ_X and variance σ_X^2

Y: another (independent) measurement with another device with mean μ_Y and variance σ_Y^2

$Z = cX+(1-c)Y$

Can we find the value of c that minimizes $\mathrm{Var}(Z)$?

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Linear Combinations of Measurements

(X_1, X_2): a possibly correlated sample with mean μ_i and variance σ_i^2

$Y = c_1X_1+ c_2X_2$: function of (X_1, X_2), where c_i's are constants

$E(Y)$ and $\mathrm{Var}(Y)$: ?

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a possibly correlated sample with mean μ_i and variance σ_i^2

$Y = \sum_{i=1}^{n} c_i X_i$: function of $(X_1, X_2, \cdots, X_n)$, where c_i's are constants

$E(Y)$ and $\mathrm{Var}(Y)$: ?

$$E(Y) = E(\sum_{i=1}^{n} c_i X_i) = \sum_{i=1}^{n} c_i \mu_i$$

$$Var(Y) = Var(\sum_{i=1}^{n} c_i X_i) = \sum_{i=1}^{n} c_i^2 \sigma_i^2 + 2 \sum_{i=1}^{n-1} \sum_{j=i+1}^{n} c_i c_j Cov(X_i, X_j)$$

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Linear Combinations of Measurements

$(X_1, X_2, \cdots, X_n)$: a possible correlated sample with mean μ_i and variance σ_i^2

$Y = \sum_{i=1}^{n} c_i X_i$: function of $(X_1, X_2, \cdots, X_n)$, where c_i's are constants

An upper bound on σ_Y : ?

$$\sigma_Y \leq \sum_{i=1}^{n} |c_i| \sigma_i$$

See your book for the details of the derivation.

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Example

A surveyor is measuring the perimeter of a rectangular lot. He measures two adjacent sides to be 50.11 ± 0.05 m and 75.21 ± 0.08 m. These measurements are not necessarily independent. Find a conservative estimate of the uncertainty in the perimeter of the lot.

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Linear Combinations of Measurements

Two independent random samples are observed:

$(X_1, X_2, \cdots, X_n)$: a random sample with common mean μ_X and variance σ_X^2

$(Y_1, Y_2, \cdots, Y_n)$: a random sample with common mean μ_Y and variance σ_Y^2

$Z = a\bar{X} + b\bar{Y}$

$E(Z)$ and $\mathrm{Var}(Z)$: ?

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Nonlinear Functions of Measurements

See your book if interested. We skip this section in this course.

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Summary

We discussed measurement error.

Main concepts:

- Different contributions to measurement error: bias and uncertainty
- Linear combinations of measurements:
  - Independent measurements with equal and different uncertainties
  - Dependent measurements

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Class exercise

Consider the random variables X and Y with the following joint probability mass function:

$P_{X,Y}(1,1) = P_{X,Y}(0,0) = \frac{3}{8}$; $P_{X,Y}(0,1) = P_{X,Y}(1,0) = \frac{1}{8}$

a) (E[X], E[Y])=?

b) (Var(X), Var(Y))=?

c) Cov(X,Y)=?

d) Var(2X+3Y)=?

$$Var(X) = E[(X - \mu_X)^2] = E[X^2] - (\mu_X)^2$$
$$Cov(X,Y) = E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X \mu_Y$$
$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2abCov(X,Y)$$

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